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A phylogenetic study of *Saxifraga* sect. *Saxifraga* (Saxifragaceae) based on nrDNA ITS sequences

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Abstract. Parsimony analyses of 54 nrDNA ITS (Internal Transcribed Spacer) sequences of *Saxifraga* sect. *Saxifraga* were performed. In addition to some unresolved clades, there is strong disagreement between the ITS phylogeny and previous classifications based primarily on morphology. The extensive cytological instability of sect. *Saxifraga* prevents previous cytotaxonomical results from resolving the incongruence between molecular and morphological data. Dissimilar topologies between chloroplast (*mat*K) and nuclear (ITS) trees for eight species of sect. *Saxifraga* suggest that gene trees and the true species tree are not coincident. Recent and mid-term reticulation is proposed as an explanation for the incongruence between morphological, cytological, organellar, and nuclear data. Homogenization in multigene families, such as the ITS region, via concerted evolution may be the key to the interpretation of results based on ITS sequences within sect. *Saxifraga*. The use of organellar genes in a larger sample should help to determine whether extensive reticulation occurs in sect. *Saxifraga*, as has been documented in various genera of Saxifragaceae.

Key words: Saxifragaceae, *Saxifraga*, sect. *Saxifraga*, ITS sequences, classification, reticulation

Saxifraga L. consists of ~500 species distributed in Asia, Europe, North and South America, and northern Africa. The primarily European sect. *Saxifraga* of about 70 species (Webb and Gornall 1989) is one of the infrageneric groups to encompass a broad range of morphological, cytological, ecological, and life cycle diversity in the genus. Examples of variation in morphological characters are: a habit that ranges from compact cushions to single-stemmed herbs; roots that range from axonomorphous to fasciculate emerging from bulblets; leaves that range from compound to entire; flowering stems that develop from terminal or axillary rosette buds; flowers that are solitary or arranged in many-flowered inflorescences; an ovary position that varies from semi-inferior to inferior; and petals that are glabrous or hairy (Webb and Gornall 1989, Vargas 1997) (Table 1). On the other hand, a distinct set of features defines sect. *Saxifraga*: uniseriate haris (Gornall 1986); pollen that is characterised primarily by a continuous granular-spinulose tectum (Ferguson and Webb 1970); leaves alternate, usually lobed or divided to some extent, without lime-secreting hydathodes; an ovary position that ranges from semi-inferior to almost completely inferior; petals usually white, without spots (Webb and Gornall 1989, Webb 1993). The perennial habit is found in all *Saxifraga* species, except for one annual (*S. tridactylites*) and three biennial (*S. adscendens*, *S. blavii*, and *S. osloensis*) species in sect. *Saxifraga*.

Table 1. Distinctive characters and character states on which the classification of *Saxifraga* sect. *Saxifraga* has primarily been based

Character	Character states
1 Underground bulblets	(0) Absent; (1) Present
2 Leaf	(0) Entire to lobed; (1) Ternate to compound
3 Summer-dormant buds	(0) Absent; (1) Present
4 Leaf furrows	(0) Absent; (1) Present
5 Leaf pubescent	(0) Present; (1) Absent
6 Flowering stems	(0) Terminal; (1) Axillary
7 Hairiness of petals	(0) Absent; (1) Present
8 Petal colour	(0) Fully white; (1) Other
9 Seed ornamentation	(0) Papillosae; (1) Papillosae + tuberculatae; (2) Smooth

A monocarpic species (*S. latepetiolata*) is also included in this section. Aneuploidy (dysploidy) and polyploidy (allopolyploidy) have been described as mechanisms to be responsible for chromosome number variation among and within species in the group (Knaben 1954, Löve and Löve 1961, Federov 1974, Moore 1982, Vargas and Nieto Feliner 1995). Chromosome numbers ranging from 2n = 20 in *S. exarata* (Favarger 1965) to 2n = 210 in *S. androsacea* (Küpfer 1972) illustrate the complex cytology of sect. *Saxifraga*, in contrast with the comparative uniformity of other sections: sect. *Cymbalaria* Griesb. (2n = 18), sect. *Cotylea* Tausch (2n = 22), sect. *Gymnopera* D. Don (2n = 28), sect. *Porphyrion* Tausch (2n = 26), sect. *Ligulatae* Haw. (2n = 28). The marked endemicity of many European species in sect. *Saxifraga* to small mountain ranges, along with their specificity for particular rock substrates either in the Eurosiberian or Mediterranean regions account for their wide use as character species in plant community studies (Peinado and Rivas-Martínez 1987).

The morphological and biosystematic information summarized above have recently been used to circumscribe seemingly natural groups within sect. *Saxifraga*. Despite the considerable knowledge of the taxonomy of this assemblage, there is no classification consensus either between authors or between subsequent treatments published by the same author. Table 2 shows four different classifications proposed for the European species in the last 12 years. In addition to the placement of some species into disparate supraspecific taxa, the major classification differences among the four treatments are: recognition of series in subsect. *Saxifraga* or not, placement of series *Gemmiferae* into either subsect. *Saxifraga* or subsect. *Triplinervium*, acceptance of series in subsect. *Triplinervium* or not. The use of a combination of non-unique morphological characters to define some supraspecific taxa raises the possibility of non-monophyletic groups in sect. *Saxifraga*. Although chloroplast and nuclear evidence for polyphyly of the genus *Saxifraga* have been provided (Soltis et al. 1993, Soltis and Soltis 1997), a phylogeny of Saxifragaceae s. s. based on *mat*K sequences contains a clade with only sect. *Saxifraga* species (Soltis el al. 1996a). Preliminary molecular studies based on nuclear ribosomal DNA ITS (Internal Transcribed Spacer) sequences point to non-monophyletic groups in sect. *Saxifraga* (Vargas et al. 1999).

To assess natural groups within sect. *Saxifraga* as a whole, ITS sequence data of 54 species representing all the subsections and series are analyzed. A higher nucleotide substitution rate and informative character number in ITS sequences (Vargas et al. 1999) than in *rbc*L or *mat*K sequences (Soltis et al. 1996a, Brochmann et al. 1996, Brochmann et al. 1998) lead to the evaluation of nuclear ribosomal DNA variation. The phylogenetic use of ITS sequences is chosen for re-examining morphological and chromosome evolution in sect. *Saxifraga*.

Materials and methods

Plant materials. Pooled leaf tissue of one to five individuals per population from 35 species was sampled to represent all the subsections and series of sect. *Saxifraga*, including the type species (Tables 2, 3). Voucher specimens are deposited in

Table 2. Classifications of European (Webb 1987, Webb and Gornall 1989, Webb 1993), Macaronesian (Vargas 1991), and Iberian (Vargas 1997) species of *Saxifraga* sect. *Saxifraga* published in the last 12 years. Sequences of species in bold were used in this study. Asterisk (*) after species names indicates updated names. Numbers before supraspecific names refer to numbers in Fig. 1

Webb 1987	Webb and Gornall 1989	Webb 1993	Vargas 1991, 1997
subsect. Saxifraga	subsect. Saxifraga	subsect. Saxifraga	subsect. Saxifraga
S. granulata L.	**1** series Saxifraga	**1** series Saxifraga	**1** series Saxifraga
S. corsica (Ser.) Gren. & Godron	***S. granulata***	***S. granulata***	***S. granulata***
S. haenseleri Boiss. & Reut.	***S. corsica***	***S. corsica***	***S. corsica*** (incl. ***S. cossoniana*** Godr.)
S. dichotoma Willd.	***S. haenseleri***	***S. haenseleri***	***S. haenseleri***
S. carpetana Boiss. & Reut.	***S. dichotoma***	***S. dichotoma***	***S. dichotoma***
S. cintrana Kuzinsky ex Willk	***S. carpetana*** (incl. ***S. graeca*** Boiss. & Heldr.)	***S. carpetana***	***S. carpetana***
S. bulbifera L.		***S. cintrana***	***S. cintrana***
S. gemmulosa Boiss.	***S. cintrana***	***S. bulbifera***	**2** series Biternatae
S. bourgeana Boiss. & Reut.	***S. bulbifera***	***S. gemmulosa***	***S. gemmulosa***
S. biternata Boiss.	**2** series Biternatae (Engl. & Irmsch.) Gornall	***S. bourgeana***	***S. bourgeana***
subsect. Triplinervium (Gaudin) Gornall		***S. biternata***	***S. biternata***
	S. gemmulosa	**2** series Gemmiferae	**3** series Gemmiferae
3 series Gemmiferae (Willk.) Pawlowska	***S. bourgeana***	***S. globulifera***	***S. globulifera***
	S. biternata	***S. reuteriana***	***S. reuteriana***
S. globulifera Desf.	subsect. Triplinervium	***S. erioblasta***	***S. erioblasta***
S. reuteriana Boiss.	**3** series Gemmiferae	***S. rigoi***	***S. rigoi***
S. erioblasta Boiss. & Reut.	***S. globulifera***	***S. conifera***	*S. trabutiana* Engl. & Irmsch.
S. rigoi Porta	***S. reuteriana***	***S. fragosoi***	
S. conifera Coss. & Durieu	***S. erioblasta***	***S. hypnoides***	***S. conifera***
S. fragosoi* Sennen	***S. rigoi***	subsect. Triplinervium	***S. fragosoi***
S. hypnoides L.	***S. conifera***	***S. rosacea***	subsect. Triplinervium
11 series Cespitosae (Reichenb.) Pawlowska	***S. fragosoi***	***S. cespitosa***	**4** series Axilliflorae
	S. hypnoides	***S. pubescens***	***S. praetermissa***
S. rosacea Moench	**11** series Cespitosae	***S. nevadensis***	**5** series Aquaticae
S. cespitosa L.	***S. rosacea*** (incl. ***S. hartii*** D. A. Webb)	***S. cebennensis***	***S. aquatica***
S. pubescens Pourr.		***S. exarata***	**6** series Ceratophyllae
S. nevadensis Boiss.	***S. cespitosa***	***S. hariotii***	***S. babiana***
S. cebennensis Rouy & Camus	***S. pubescens***	***S. praetermissa***	***S. trifurcata***
S. exarata Vill.	***S. nevadensis***	*S. wahlenbergii*	***S. cuneata***
S. hariotii Luizet & Soulié	***S. cebennensis***	*S. pedemontana*	***S. maderensis*** D. Don
4 series Axilliflorae (Willk.) S. Pawlowska	***S. exarata***	***S. babiana***	***S. portosanctana*** Boiss.
	S. hariotii	***S. trifurcata***	**7** series Pentadactyles (Lázaro Ibiza) P. Vargas
S. praetermissa D. A. Webb	**4** series Axilliflorae	***S. cuneata***	

Table 2. (Continued)

Webb 1987	Webb and Gornall 1989	Webb 1993	Vargas 1991, 1997
S. wahlenbergii Ball	***S. praetermissa***	***S. vayredana***	***S. pubescens***
6 series Ceratophyllae (Haw.) Pawlowska	*S. wahlenbergii*	***S. intricata***	***S. nevadensis***
S. pedemontana All.	**6** series Ceratophyllae	***S. pentadactylis***	***S. harioti***
S. babiana T. E. Díaz & Fern. Prieto	*S. pedemontana*	***S. fragilis***	***S. vayredana***
S. trifurcata Schrad.	***S. babiana***	***S. camposii***	***S. intricata***
S. cuneata Willd.	***S. trifurcata***	***S. geranioides***	***S. pentadactylis***
S. vayredana Luizet	***S. cuneata***	***S. canaliculata***	***S. fragilis***
S. intricata* Lapeyr	***S. vayredana***	***S. moncayensis***	***S. camposii***
S. pentadactylis Lapeyr.	***S. intricata***	***S. aquatica***	***S. geranioides***
S. fragilis Schrank	***S. pentadactylis***	*S. irrigua*	***S. genesiana*** P. Vargas
S. camposii Boiss. & Reut.	***S. fragilis***	***S. latepetiolata***	***S. canaliculata***
S. geranioides L.	***S. camposii***	*S. petraea*	***S. moncayensis***
S. canaliculata Boiss. & Reut.	***S. geranioides***	*S. berica*	***S. losae*** *Sennen*
S. moncayensis D. A. Webb	***S. canaliculata***	***S. arachnoidea***	***S. latepetiolata***
7 series Aquaticae (Engl.) Pawlowska	***S. moncayensis***	*S. paradoxa*	**9** subsect. Holophyllae
S. aquatica Lapeyr.	**5** series Aquaticae (Engl. & Irmsch.) Pawlowska		***S. moschata*** Wulfen
S. irrigua Bieb.	***S. aquatica***		***S. androsacea***
S. latepetiolata Willk.	*S. irrigua*		***S. sedoides***
	S. latepetiolata		
8 series Arachnoideae	**8** series Arachnoideae	**9** subsect. Holophyllae	**10** subsect. Tridactylites
S. petraea L.	*S. petraea*	***S. androsacea***	***S. tridactylites***
S. berica (Béguinot) D. A. Webb	*S. berica*	***S. depressa***	***S. adscendens***
S. arachnoidea Sternb.	***S. arachnoidea***	*S. italica*	
S. paradoxa Sternb.	*S. paradoxa*	S. *seguieri*	
9 subsect. Holophyllae (Engl.) Engl. & Irmsch.	**9** subsect. Holophyllae	*S. muscoides*	
	S. androsacea	***S. fachinii***	
S. androsacea L.	***S. depressa***	*S. presolanensis*	
S. depressa Sternb.	*S. italica*	***S. sedoides***	
S. italica D. A. Webb	*S. seguieri*	***S. aphylla***	
S. seguieri Spregel	*S. muscoides*	*S. glabella*	
S. muscoides All.	***S. fachinii***	*S. tenella*	
S. fachinii Koch	*S. presolanensis*	**10** subsect. Tridactylites	
S. presolanensis Engl.	***S. sedoides***	***S. tridactylites***	
S. sedoides L.	***S. aphylla***	***S. adscendens***	
S. aphylla Sternb.	*S. glabella*	*S. blavii*	
S. glabella Bertol.	*S. tenella*	***S. osloensis***	
S. tenella Wulfen	**10** subsect. Tridactylites		
10 subsect. Tridactylites (Haw.) Gornall	***S. tridactylites***		
S. tridactylites L.	***S. adscendens***		
S. adscendens L.	*S. blavii*		
S. blavii Engl.	***S. osloensis***		
S. osloensis Knaben			

Table 3. Taxonomic list of plant material from representative populations of *Saxifraga* species from which ITS sequences were obtained in this study, with localities, collection references, and GenBank accession numbers given. Abbreviations: **JACA** Herbarium of the Pyrenees Institute of Jaca (Huesca, Spain); **MA** Herbarium of the Royal Botanical Garden of Madrid, **RNG** Herbarium of the University of Reading; **PV** Pablo Vargas

Taxon	Locality and collection reference	Accession no.
S. androsacea	Switzerland, Valais, S. Castroviejo 11144 (MA)	AF261159
S. aphylla	Austria, Innsbruck, D. Gómez (JACA)	AF261160
S. aquatica	Spain, Huesca, Bielsa, 34-97PV (MA)	AF261161
S. arachnoidea	Italy, Lombardía, Riva del Garda (RNG)	AF261162
S. biternata	Spain, Málaga, Antequera, 8-97PV (MA)	AF261163
S. bourgeana	Spain, Cádiz, Grazalema, 7-97PV (MA)	AF261164-AF261165
S. bulbifera	Italy, Fiuminata, Conti (MA)	AF261166
S. canaliculata	Spain, León, Barrios de Luna, 2991PV (MA)	AF261167
S. carpetana	Spain, Avila, Solosancho, 12-97PV (MA)	AF261168
S. cebennensis	France, Gorge de Tarn, 51-97PV (MA)	AF261169
S. cespitosa	Canada, Newfoundland, W. Baker 769 (RNG)	AF261170
S. cintrana	Portugal, Sierra de Montejunto, 2515PV (MA)	AF261171
S. cossoniana	Spain, Alicante, Vall de Laguart, 10, G. López 732 (MA)	AF261172
S. depressa	Italy, Dolomites, Cazione, 70-97PV (MA)	AF261173
S. dichotoma	Spain, Toledo, Hinojosa de San Vicente (MA)	AF261174
S. fachinii	Italy, Dolomites, passo di Sella, 73-97PV (MA)	AF261175
S. fragilis	Spain, Castellón, Morella, 3147PV (MA)	AF261176
S. gemmulosa	Spain, Málaga, Sierra Bermeja, 2-97PV (MA)	AF261177
S. genesiana	Spain, Barcelona, Sierra del Montseny, A. Bombí (MA)	AF261178
S. geranioides	Andorra, El serrat, C. Aedo 2099 (MA)	AF261192
S. graeca	Greece, Olimbos, Stauros, Richards (RNG)	AF261179
S. haenseleri	Spain, Albacete, Sierra de Alcaraz, G. Nieto 1297 (MA)	AF261180
S. hariotii	France, Atlantic Pyrenees (JACA)	AF261181
S. hederacea	Libya, Wadi Kouf, Davis 50203 (RNG)	AF261182
S. latepetiolata	Spain, Cuenca, Ciudad Encantada (MA)	AF261183
S. moschata	Spain, Pyrenees, Huesca, Portalet, 25-97PV (MA)	AF261184
S. nevadensis	Spain, Sierra Nevada, MA508846	AF261185
S. praetermissa	Spain, Cantabria, Picu Uriellu, 3250PV (MA)	AF261186
S. pubescens	France, Hautes Pyrenees (JACA)	AF261187-AF261188
S. hartii	cm 457 (kew1969-51294)	AF261189
S. rosacea	RNG	AF261190
S. sedoides	Italy, Dolomites, Marmolada Massif, 60-97PV (MA)	AF261192

the MA (Royal Botanical Garden of Madrid) and RNG (University of Reading) herbaria.

DNA extraction, gene amplification, and sequencing. Total DNA was extracted from silica dried material and herbarium specimens, according to the CTAB method of Doyle and Doyle (1987), as modified by Loockerman and Jansen (1996) using two alcohol precipitations. PCR amplifications were performed using the external oligonucleotide primers 17SE (5′-ACGAATTCATGGTCCGGTGAAGTGTTCG-3′) and 26SE (5′-TAGAATTCCCCGGTTCGCTCGCCGTTAC-3′) (Sun et al. 1994) or c28kj (5′-TTGGACGGAATTTACCGCCCG-3′, designed by K. W. Cullings, San Francisco State University) and LEU1 (5′-GTCCACTGAACCTTATCATTTAG-3′, designed by L. E. Urbatsch, Louisiana State University). Amplified products were cleaned using QIAGEN QIAquick TM PCR Purification kit (QIAGEN Inc., Chatsworth, CA) following protocols provided by the manufacturer. Cleaned products were then directly sequenced using the ABI PRISM Dye

Terminator Cycle Sequencing Ready Reaction Kit with AmpliTaq DNA Polymerase, FS (Perkin-Elmer Corp.). For cycle sequencing, the primers ITS4 and ITS5 were mostly used, but primers ITS2 and ITS3 (White et al. 1990) were necessary for sequencing *S. bourgeana* and *S. pubescens*. Unincorporated dye terminators were removed using sodium acetate and absolute alcohol precipitations. Samples were loaded into a 5.75 or 7.00% polyacrylamide gel, analyzed on ABI 373A and ABI 377 DNA Sequencers, and finally, data were collected on a Macintosh platform. Sequencing data were edited and assembled using the program "Sequencer" (Gene Codes Corporation, Miami).

Analysis. Thirty-one new ITS sequences plus 23 ITS sequences withdrawn from Brochmann et al. (1996) and GenBank (AJ233858–AJ233864; AJ233874–AJ233876; AJ233878–AJ233880; AJ233882–AJ233884; AJ133030) (Vargas et al. 1999) were used as the ingroup, which represent ~75% of the species in sect. *Saxifraga* (Webb and Gornall 1989) (Table 2). DNA extraction, gene amplification, and sequencing for the 20 GenBank sequences are given in Vargas et al. (1999) and Vargas (in press). *Saxifraga hederacea* L. (sect. *Cymbalaria*) and *S. spathularis* Brot. (sect. *Gymnopera*) were chosen as the outgroup based on previous classsifications (Webb and Gornall 1989) and *mat*K results (Soltis et al. 1996a). The 31 new ITS sequences were aligned by visual inspection into the sequence matrix used for previous analyses (Vargas et al. 1999). Inferred gaps were treated as missing data. Equivocal sites were coded as polymorphic using IUPAC ambiguity symbols. Phylogenetic analyses were conducted using Fitch parsimony (as implemented in test version beta 4.0b2 PAUP*, written by D. L. Swofford, Smithsonian Institution) with equal weighting of all characters and of transitions/transversions. Heuristic searches were replicated 100 times with random taxon-addition sequences, tree bisection-reconnection (TBR) branch swapping, and with the options MULPARS and STEEPEST DESCENT in effect. Reliability of lineages was assessed by bootstrapping (100 resamplings of data), using the heuristic search strategy indicated above.

Two more analyses were performed using the above options in 4.0b2 PAUP* in order to compare chloroplast and nuclear tree topologies. The first one contains eight *mat*K sequences of sect. *Saxifraga* already published plus those of *S. hederacea* and *S. spathularis* (outgroup) (Soltis et al. 1996a). The second one includes only the ITS sequences of those species used in the analysis of *mat*K sequences.

Results

ITS sequence variation. The ITS region (ITS-1, 5.8S, and ITS-2) in sect. *Saxifraga* is 680 to 667 bp in length (280 to 264 in ITS-1, 164–165 bp in 5.8S, and 239 to 231 in ITS-2). *Saxifraga spathularis* has the shortest sequence (667 bp), being the longest sequence in *S. conifera*, *S. camposii*, and *S. carpetana* subsp. *graeca* (680 bp). The aligned sequence matrix generated 706 characters, 227 of which were potentially informative. Alignment of the ITS sequences of *Saxifraga* required introduction of gaps: 53 from length differences with the outgroup (*S. hederacea* and *S. spathularis*) and 28 from length differences among sect. *Saxifraga* species. Within sect. *Saxifraga*, pairwise comparisons of the ITS-region sequences using the Kimura 2-parameter model of sequence evolution yielded divergences ranging from 18.00% (between *S. adscendens* and *S. corsica* subsp. *cossoniana*) to 0.32% (between *S. pubescens* and *S. hariotii*). The highest divergence value for ITS-1 was found between *S. maderensis* and *S. tridactylites* (22.76%), and the highest divergence for ITS-2 was found between *S. corsica* subsp. *cossoniana* and *S. adscendens* (17.23%), including 44 missing data from *S. tridactylites*, *S. adscendens*, and *S. osloensis* each (Brochmann et al. 1996). All these values are somewhat higher than those already reported for sect. *Saxifraga* (Vargas et al. 1999). Infraspecific ITS variation showed the highest divergence between *S. carpetana* Boiss. & Reuter subspp. *carpetana* and *gracea* (Boiss. & Heldr.) D. A. Webb (7.20%) and the lowest between *S. rosacea* Moench subspp. *rosacea* and *hartii* (D. A. Webb) D. A. Webb (0.89%). Specific status has also been recognized for these four taxa (see Table 2).

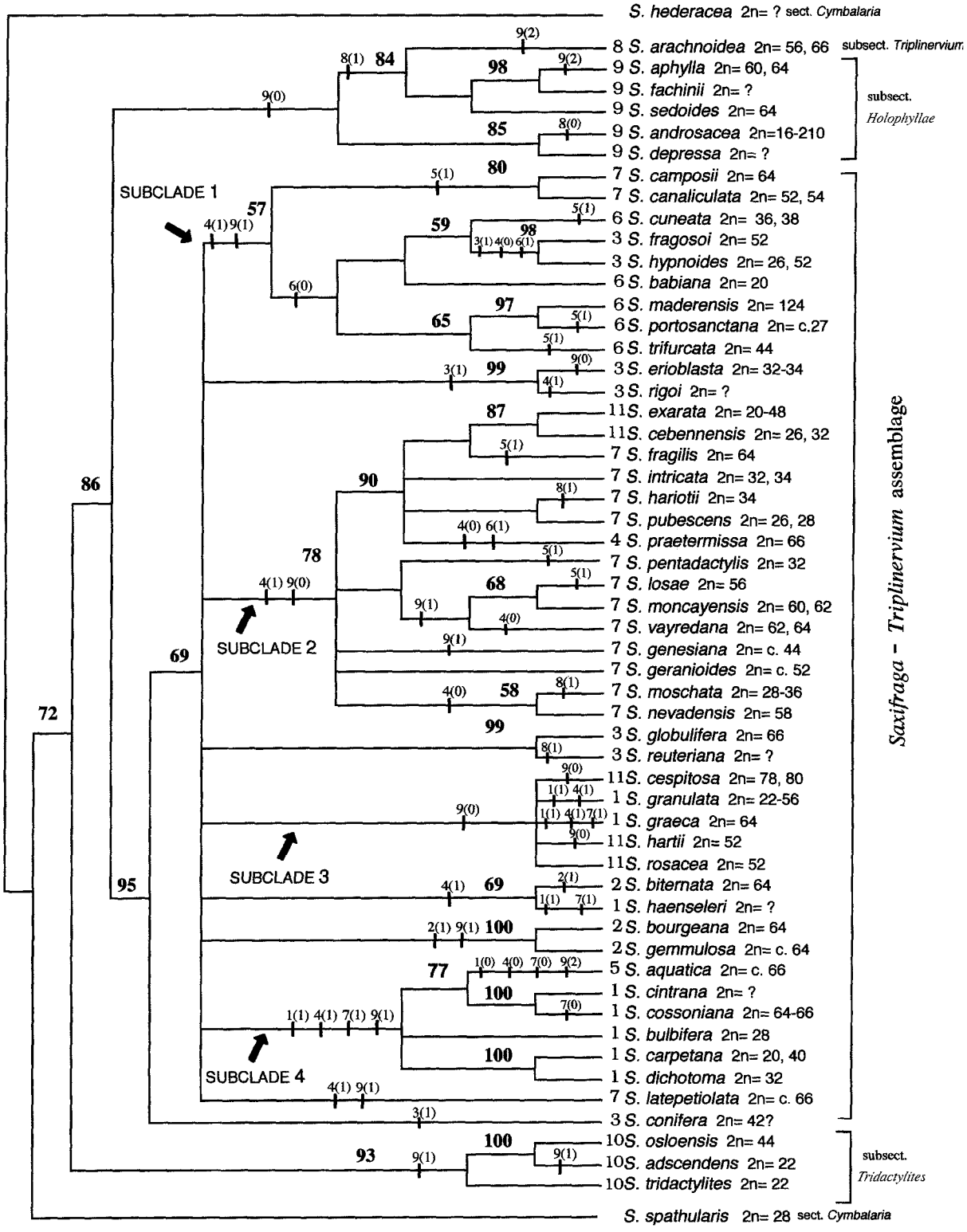
Phylogenetic analysis of ITS sequences. The heuristic searches found 14,304 minimum-

length trees of 838 steps (CI: 0.62, including uninformative characters; RI: 0.70). No long ITS character branches were observed. The resulting semi-strict consensus using *S. hederacea* as the outgroup gives rise to a clade with the 54 sequences of sect. *Saxifraga* (72% bootstrap) (Fig. 1). Within this sect. *Saxifraga* clade, one basal, well-defined clade (93% bootstrap) is formed by the three species of subsect. *Tridactylites*, sister to a clade that includes all other species (86% bootstrap). The next, basal grouping of the five species of subsect. *Holophyllae* plus *S. arachnoidea* (subsect. *Saxifraga*) is, in turn, weakly supported (< 50% bootstrap). The remaining 45 species of subsect. *Saxifraga* and subsect. *Triplinervium* form a well-supported clade (95% bootstrap). Apart from the basal position of *S. conifera* within this subsect. *Saxifraga-Triplinervium* clade, there is no basal resolution due to the formation of a polytomy by nine subclades (Fig. 1). Subclades with more than two species are identified as follows: subclade 1 of *S. camposii-S. canaliculata-S. cuneata-S. fragosoi-S. hypnoides-S. babiana-S. maderensis-S. portosanctana-S. trifurcata* (57% bootstrap); subclade 2 of *S. exarata-S. cebennensis-S. fragilis-S. intricata-S. hariotii-S. pubescens-S. praetermissa-S. pentadactylis-S. losae-S. moncayensis-S. vayredana-S.genesiana-S. geranioides-S. moschata-S. nevadensis* (78% bootstrap); subclade 3 of *S. cespitosa-S. granulata-S. graeca-S. hartii-S. rosacea* (< 50% bootstrap); and subclade 4 of *S. aquatica-S. cintrana-S. cossoniana-S. bulbifera-S. carpetana-S. dichotoma* (< 50% bootstrap). Four species-pair subclades can also be recognized in the subsect. *Saxifraga-Triplinervium* clade: *S. erioblasta-S. rigoi* (99% bootstrap), *S. globulifera-S. reuteriana* (99% bootstrap), *S. biternata-S. haenseleri* (69% bootstrap), and *S. bourgeana-S. gemmulosa* (100% bootstrap).

Phylogenetic analysis of *mat*K and ITS sequences. ITS and *mat*K sequences are available for eight species of sect. *Saxifraga*: *S. adscendens*, *S. osloensis*, *S. arachnoidea*, *S. fragilis*, *S. latepetiolata*, *S. cuneata*, *S. cespitosa*, and *S. exarata*. The heuristic searches found one single most-parsimonious tree of 350 steps (CI: 0.84, including uninformative characters; RI: 0.71) using the eight ITS-region sequences and one single most-parsimonious tree of 150 steps (CI: 0.89, including uninformative characters; RI: 0.79) using the eight *mat*K sequences. Within sect. *Saxifraga*, *S. fragilis* is the most basal species in the *mat*K tree, in contrast to the *S. osloensis-S. adscendens* clade in the ITS tree (Fig. 2). Neither are terminal clades congruent between both topologies. Well-supported clades in the ITS tree, such as *S. cuneata-S. cespitosa-S. latepetiolata* (89% bootstrap) or *S. exarata-S. fragilis* (100% bootstrap), are not found in the *mat*K tree.

Discussion

Phylogenetic resolution. The semi-strict consensus tree (Fig. 1) retrieved from the 54 ITS sequences of sect. *Saxifraga* and ITS sequences of *S. hederacea* and *S. spathularis* (outgroup) does not disagree with the monophyly of sect. *Saxifraga* found using *mat*K sequences (Soltis et al. 1996). Within sect. *Saxifraga*, the only well-defined subsection that is shown to be monophyletic is subsect. *Tridactylites* (93% bootstrap), whereas species of subsect. *Holophyllae* plus *S. arachnoidea* (subsect. *Triplinervium*) are grouped with weak support (< 50% bootstrap). The remaining 45 species of subsect. *Triplinervium* and *Saxifraga* form a monophyletic group (95% bootstrap) poorly resolved at the basal level in the semi-strict consensus tree (Fig. 1). Despite a high number of potentially informative characters (125 in ITS-1, 6 in 5.8S, and 96 in ITS-2), a broad polytomy of nine subclades may indicate strong incongruence among ITS characters, thus preventing the reconstruction of a single phylogeny. In addition to four species-pair subclades and. *S. latepetiolata*, there are four subclades with low and medium support (Fig. 1): subclade 1, which has been discussed elsewhere (Vargas et al. 1999); subclade 2, with species primarily distributed in northern Spain and southern France; subclade 3, which


S. hederacea 2n= ? sect. Cymbalaria
8 S. arachnoidea 2n= 56, 66
subsect. Triplinervium
9 S. aphylla 2n= 60, 64
9 S. fachinii 2n= ?
9 S. sedoides 2n= 64
9 S. androsacea 2n=16-210
9 S. depressa 2n= ?
subsect. Holophyllae
7 S. camposii 2n= 64
7 S. canaliculata 2n= 52, 54
6 S. cuneata 2n= 36, 38
3 S. fragosoi 2n= 52
3 S. hypnoides 2n= 26, 52
6 S. babiana 2n= 20
6 S. maderensis 2n= 124
6 S. portosanctana 2n= c.27
6 S. trifurcata 2n= 44
3 S. erioblasta 2n= 32-34
3 S. rigoi 2n= ?
11 S. exarata 2n= 20-48
11 S. cebennensis 2n= 26, 32
7 S. fragilis 2n= 64
7 S. intricata 2n= 32, 34
7 S. hariotii 2n= 34
7 S. pubescens 2n= 26, 28
4 S. praetermissa 2n= 66
7 S. pentadactylis 2n= 32
7 S. losae 2n= 56
7 S. moncayensis 2n= 60, 62
7 S. vayredana 2n= 62, 64
7 S. genesiana 2n= c. 44
7 S. geranioides 2n= c. 52
7 S. moschata 2n= 28-36
7 S. nevadensis 2n= 58
3 S. globulifera 2n= 66
3 S. reuteriana 2n= ?
11 S. cespitosa 2n= 78, 80
1 S. granulata 2n= 22-56
1 S. graeca 2n= 64
11 S. hartii 2n= 52
11 S. rosacea 2n= 52
2 S. biternata 2n= 64
1 S. haenseleri 2n= ?
2 S. bourgeana 2n= 64
2 S. gemmulosa 2n= c. 64
5 S. aquatica 2n= c. 66
1 S. cintrana 2n= ?
1 S. cossoniana 2n= 64-66
1 S. bulbifera 2n= 28
1 S. carpetana 2n= 20, 40
1 S. dichotoma 2n= 32
7 S. latepetiolata 2n= c. 66
3 S. conifera 2n= 42?
10 S. osloensis 2n= 44
10 S. adscendens 2n= 22
10 S. tridactylites 2n= 22
subsect. Tridactylites
S. spathularis 2n= 28 sect. Cymbalaria
Saxifraga - Triplinervium assemblage
SUBCLADE 1
SUBCLADE 2
SUBCLADE 3
SUBCLADE 4
84
98
85
80
57
59
98
97
65
99
87
90
78
68
69
58
99
69
100
77
100
100
100
93
86
72
95

includes the type species of *Saxifraga* and Saxifragaceae (*S. granulata*); and subclade 4, which includes species with underground bulbils, as well as *S. aquatica*.

Monophyletic groups and support of previous classifications. There is no fully supported classification in sect. *Saxifraga* using the ITS sequence data. Although the three species of subsect. *Tridactylites* form a monophyletic group, the other three subsections include intermingled species (Fig. 1). The subsect. *Holophyllae*, *Saxifraga*, and *Triplinervium* do not form monophyletic groups in the ITS tree according to any of the previous classifications. A basal clade with weak support (< 50% bootstrap) contains the five species sequenced of subsect. *Holophyllae* plus *S. arachnoidea* (subsect. *Triplinervium*), whereas the other 45 species of subsect. *Saxifraga* and *Triplinervium* form a well-defined assemblage (95% bootstrap). The relationship among groups of species within the *Saxifraga-Triplinervium* assemblage are completely unresolved at the basal level (Fig. 1), and few clades contain species exclusive to one of the two subsections: four unresolved species pairs (subsect. *Saxifraga*) and subclade 1 and subclade 2 (subsect. *Triplinervium*). The ITS phylogeny does not support the serial treatment proposed by different authors (Table 1), and polyphyly in many series of subsect. *Saxifraga* and *Triplinervium* is observed (Fig. 1).

Origin of the *Triplinervium-Saxifraga* assemblage. The basal position in the ITS tree of the clades containing the 8 species of subsect. *Tridactylites* and *Holophyllae* may imply that the 45 species belonging to the *Saxifraga-Triplinervium* assemblage are the most derived. Although the most basal species of sect. *Tridactylites* (*S. tridactylites*) is broadly distributed across Europe, the primarily alpine species of sect. *Holophyllae* may rule out the possibility of a Mediterranean origin for the *Triplinervium-Saxifraga* assemblage. Based on the ITS phylogeny, all Mediterranean species are at mid-level or terminal clades, which supports previous considerations of an arctic-alpine origin of the sect. *Saxifraga* (Webb and Gornall 1989) (but see *S. fragilis* in the *mat*K topology in Fig. 2).

Sources of discord. There is a marked incongruence between the ITS tree and previous classifications based primarily on morphological characters (Webb 1987, Webb and Gornall 1989, Webb 1993, Vargas 1997). The phylogenetic analyses suggest disparate origins of morphologies, which have been serving to classify groups of species in sect. *Saxifraga*. Striking morphological characters cannot be mapped on single clades of the ITS phylogeny of sect. *Saxifraga* (Table 1, Fig. 1): underground bulbils (series *Saxifraga*) in three subclades, ternate to compound leaves (series *Biternatae*) in two subclades, summer-dormant buds (series *Gemmiferae*) in four subclades, axillary flowering stems (series *Ceratophyllae* and *Axilliflorae*) in two non-monophyletic subclades, glandular hairy petals (in some

Fig. 1. Semi-strict consensus tree of 14,304 most parsimonious trees of 838 steps from the heuristic analysis of the ITS-region sequences in *Saxifraga* (CI: 0.62, including uninformative characters; RI: 0.70). *Saxifraga hederacea* served as outgroup. Bootstrap values > 50% are shown above tree branches. The four subclades are discussed in the text. Numbers after species names refer to chromosome number. Symbols before species names indicate immediate supraspecific classification in sect. *Saxifraga* following Vargas (1991) for Macaronesian species, Vargas (1997) for the Iberian taxa, and Webb and Gornall (1989) for the rest of European taxa: **1** series *Saxifraga*, **2** series *Biternatae*, **3** series *Gemmiferae*, **4** series *Axilliflorae*, **5** series *Aquaticae*, **6** series *Ceratophyllae*, **7** series *Pentadactyles*, **8** series *Arachnoidea*, **9** subsect. *Holophyllae*, **10** subsect. *Tridactylites*, **11** series *Cespitosae*. *Saxifraga moschata* is included in series *Pentadactyles*. Solid boxes mapped on terminal clades of the ITS tree refer to character and character states (Table 1) Abbreviations: *S. hartii* = *S. rosacea* Moench subsp. *hartii* (D. A. Webb) D. A. Webb; *S. graeca* = *S. carpetana* Boiss. & Reuter subsp. *graeca* (Boiss. & Heldr) D. A. Webb

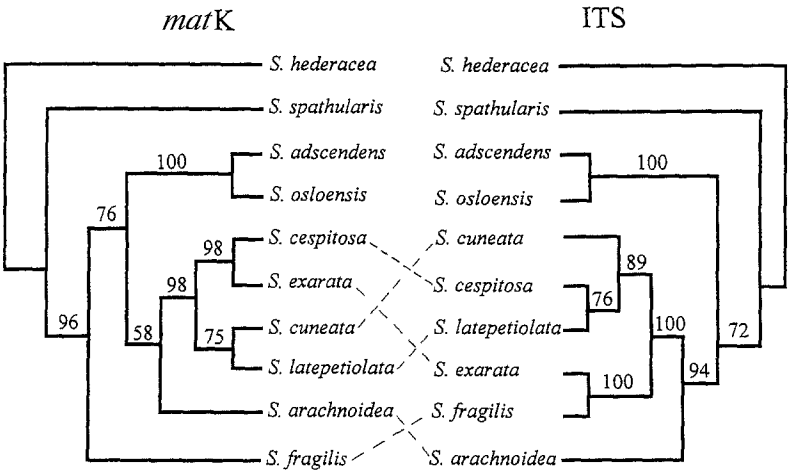


Fig. 2. Comparison of phylogenetic relationships obtained from heuristic analysis of *mat*K (left; sequences taken from Soltis et al. 1996a) and ITS (right; the present study) sequences. The two analyses involved the same set of species of *Saxifraga* sect. *Saxifraga*, plus *S. hederacea* (sect. *Cymbalaria*) and *S. spathularis* (sect. *Gymnopera*) (outgroup). Each analysis generated one single most-parsimonious tree. The *mat*K tree was of 150 steps (CI: 0.89, including uninformative characters; RI: 0.79) and the ITS tree of 350 steps (CI: 0.84, including uninformative characters; RI: 0.71). Bootstrap values above 50% are shown above tree branches

species of series *Saxifraga*) in three subclades. Other characters, such as leaf hairiness, leaf furrows, leaf acuteness, inflorescence structure, flower size, petal color, and seed ornamentation either cannot be used to support clades on the ITS tree (Fig. 1).

The fact that the gene tree obtained is completely incongruent with previous classifications indicates fundamental discordance between the ITS tree and morphological characters. Three major interpretations can be inferred from the above results: independent origin of morphological characters, lineage sorting, or reticulation (Doyle 1992). An interpretation in which every morphological character has independent origins is difficult to believe, since not only features would be affected, but also distinctive characters on which the classification of sect. *Saxifraga* has mostly been based. Neither do our results rule out the existence of lineage sorting (Neigel and Avise 1986), but the possibility of random extinction of ancestral ITS polymorphism would imply an enormous number of extinct lineages to reconcile morphological characters and the ITS phylogeny. Hybridization is the most likely explanation, for the following reasons. First, hybridization between many closely and distantly related species is common in sect. *Saxifraga* (Webb and Gornall 1989, Vargas 1997). Although F_2 generations are not easy to obtain in artificial hybridization experiments (Vargas and Nieto Feliner 1995), overcoming of genetic barriers via allopolyploidy has been detected using molecular markers (Brochmann 1996, Brochmann et al. 1998). Second, cytotaxonomical data do not support either many clades of the ITS tree (Fig. 1) or taxonomic groups (Table 2). The conjunction of dysploidy and allopolyploidy (hybridization) could explain most of the cytological instability (Vargas and Nieto Feliner 1995, Brochmann 1996, Brochmann et al. 1998) and the discordance between ITS, previous classifications, and cytotaxonomy. Third, discord between ITS and chloroplast topologies has already been determined for genera of Saxifragaceae, and interpreted as the result of ancient hybridization and chloroplast capture. For example, reticulation has been inferred not only between closely related species of *Saxifraga* (*S. rivularis* and *S. cernua*, sect. *Mesogyne*; *S. adscendens* and *S. tridactylites*, sect. *Saxifraga*) (Brochmann et al. 1998, Broch-

mann et al. 1996), but also between *Boykinia* and *Suksdorfia* (Soltis et al. 1996b) and within *Heuchera* (Soltis and Kuzoff 1995). To assess whether chloroplast capture may be responsible for some of the discordance in sect. *Saxifraga*, we have compared *mat*K (Soltis et al. 1996a) and ITS tree topologies for eight species (Fig. 2). Although only a small sample from eight species has been analyzed, discord between chloroplast DNA and ITS trees suggests that chloroplast capture has occurred in sect. *Saxifraga* to some extent.

Therefore, results of previous molecular studies indicate that reticulation has been a common evolutionary mechanism acting in many lineages of Saxifragaceae s. s., including ancient (Soltis and Kuzoff 1995, Soltis et al. 1996b) and recent (Brochmann et al. 1996, Brochmann et al. 1998) hybridization events. Molecular, cytological, and morphological evidence discussed in this study suggest that recent and mid-term reticulation may have also played an important role in the evolution of sect. *Saxifraga*. The way in which the ITS region evolves via concerted evolution (Arnheim 1983, Hillis et al. 1991) may have obscured the true species phylogeny. The use of ITS sequences alone may provide phylogenetically unreliable data when hybridization is clearly documented using other molecular markers (Brochmann et al. 1998). However, the large number of potentially informative characters at the population level encourage the use of ITS sequences in those study cases where there is no evidence of reticulation (Vargas et al. 1999, Vargas 1999).

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